

U.S. Department of Agriculture (USDA)



Attachment 1 Performance Work Statement and QASP

Analysis of On-Farm Soil Samples
USDA Columbia MO

General Information

1.0 Background and Purpose: ARS is enhancing the Soil Health Assessment Protocol and Evaluation (SHAPE) farmer-oriented decision tool and building a national soil health database for the USDA. Benchmarking and monitoring soil health is essential for farmers and ranchers to optimize the productivity and profitability of their soil resources. Farmers, including corn, soybean, cotton, citrus, wheat, and livestock producers, are participating across all 50 states, and will receive a personalized soil health report of their fields.

2.0 Scope: The Prime Contractor shall provide Analysis of On-Farm Soil Samples of approximately 10,000 samples collected from farmer participants across the US, following approved laboratory methods, to the USDA ARS Cropping Systems and Water Quality Research Government Technical Monitor as specified in this Performance Work Statement and QASP. The contractor shall be responsible for furnishing personnel, materials, equipment, shipping, facilities, and otherwise accomplish all things necessary for or incidental to the performance of the work set forth therein. This is a firm fixed price, performance-based requirement, the labor mix of employees to perform this contract remains the discretion of the Contractor for all services.

3.0 Definitions:

3.1 CLIN – Contract Line Item, numbered. Established in the solicitation schedule, will start with “0001.”

3.2 Privity of Contract - Contract service providers shall be accountable solely to the Prime Contractor, who, in turn, is responsible to the Government.

3.3 Government Technical Monitor – USDA Employee Location representative appointed by the Contracting Officer to monitor, evaluate, and communicate to the CO the Contractor’s performance in complying with contract requirements in an efficient and timely manner.

3.4 Program Office Administrative Officer – USDA Employee Location/Program Office administrative representative identified by the Contracting Officer at point of award. The Program Office Administrative Officer may not be physically located at the facility.

3.5 IDIQ – Indefinite Delivery Indefinite Quantity. The contract clauses govern administration.

3.6 Program Year – The year period that corresponds to an IDIQ Option Year.

3.7 ‘Calendar Days’- Every single day on the calendar (12 AM to 11:59 PM), including weekends, holidays, and non-working days.

3.8 ‘Business Days’ – Standard business days 8AM to 5PM USA Central Time with holiday exceptions as noted:

Listed below are official Government holidays:

New Year’s Day January 1*

Martin L. King Birthday Third Monday in January

President's Day Third Monday in February

Memorial Day Last Monday in May

Juneteenth As Indicated

Independence Day July 4*

Labor Day First Monday in September

Columbus Day Second Monday in October

Veterans Day November 11*

Thanksgiving Day Fourth Thursday in November

Christmas Day December 25*

OR when the USDA-ARS facility is closed due to local or national emergencies, administrative closings, or similar Government-directed facility closings

*In accordance with Federal regulations, if these holidays fall on a Saturday, the holiday will be observed on Friday, if Sunday, holiday will be observed on Monday.

4.0 Period of Performance: The period of performance (PoP) of the IDIQ contract shall be for one (1) Base period of 12 months and one 12-month option period.

5.0 Hours of Service Performance: Determined by Prime Contractor.

6.0 Place of Performance: Determined by Prime Contractor.

7.0 Government Provided:

7.1 Samples for analysis have been airdried and pre-sterilized following AHPIS approved methods for domestic regulated soils.

7.2 Reference samples and batch duplications for quality control.

8.0 Invoicing and Payment:

8.1 The "Submit Invoice-to" address for USDA orders is the Department of Treasury's Invoice Processing Platform (IPP).

8.2 All invoices shall be submitted via the electronic Invoice Processing Platform. This is a mandatory requirement initiated by the U.S. Department of Treasury. More information is available at <https://www.ipp.gov>. The Department of Treasury is the point of contact for all IPP Collector account inquiries.

8.3 Payment is in arrears pursuant to the clause at FAR 52.212-4. Note FAR 52.212-4 (i)(1) Items accepted. Payment shall be made for items accepted by the Government that have been delivered to the delivery destinations set forth in this contract.

8.4 Payment will not be made for service in advance of delivery.

8.5 Invoicing shall be on a no more than monthly basis and shall include reference to the contract number.

8.6 Advance invoice approval is not required, as required approval tiers exist in IPP.

Contractor Requirements

9.0 Technical Requirements/Tasks

9.1 General:

9.1.1 The Prime Contractor's laboratory shall demonstrate historical commercial competency in each specific analytical method (2 or more years of providing the analysis commercially).

9.1.2 The Prime Contractor's laboratory shall hold a quality certification/accreditation from a recognized certification program such as the North American Proficiency Testing (NAPT) Performance Assessment Program, the Agricultural Laboratory Testing Association (ALTA) Laboratory Accreditation Program, or similar program.

9.1.3 Transportation and transfer of samples:

9.1.3.1 Shipping from the USDA ARS Cropping Systems and Water Quality Research to the contractor's laboratory and any point beyond is the responsibility the contractor and included in the per sample Firm Fixed Price. Shipping will not be added to invoices at a cost beyond the Firm Fixed Price.

9.1.3.2 The contractor is responsible for arranging sample shipments. The Prime Contractor Representative shall communicate and coordinate with the Government Technical Monitor on pickup dates and times. See "Business Days" above.

9.1.3.3 Packaging and preparation of samples is completed by the USDA.

9.1.3.3.1 Samples are packaged with approximately 50 samples per box with each box weighing approximately 60 lbs.

9.1.3.3.2 If palletized, there are a maximum of 21 boxes per pallet.

9.1.4 Analytical methods may not be substituted or replaced with conversions or calculations from other methods or procedures without bi-lateral contract modification.

9.1.5 Sample intake and processing shall be consistent throughout the contract period of performance. General NAPT standards or similar shall be met.

9.1.6 Sample analysis results are due to the Government Technical Monitor via email within 90 days of sample delivery to contractor laboratory.

9.1.6.1 The Prime Contractor Representative shall transmit the results to the Government Technical Monitor.

9.1.6.2 The clause at FAR 52.246-15 Certificate of Conformance will apply.

9.1.6.3 The Prime Contractor Representative shall provide the Certificate of Conformance with all email transmittal of analysis results.

9.2 Specific Soil Properties and Methods Required:

9.2.1 Simplified particle size by pipette or hydrometer method

9.2.2 ACE total protein (Moebius-Clune, 2016)

9.2.3 Active carbon by permanganate oxidation (Weil et al., 2003)

- 9.2.4 Aggregate stability; Yoder 1-2 mm NRCS-KSSL method (Burt 2004)
- 9.2.5 Potentially mineralizable nitrogen, anaerobic method (specify incubation time and T0)
- 9.2.6 Total nitrogen and total carbon by dry combustion (Burt, 2004)
- 9.2.7 Inorganic carbon by manometer (Burt, 2004)
- 9.2.8 Organic carbon calculated by difference from total and inorganic carbon (Burt, 2004)
- 9.2.9 Soil respiration; 4-day, with base trap or IRGA
- 9.2.10 Soil respiration; 1-day, with base trap or IRGA
- 9.2.11 Electrical conductivity (1:1, soil:water) (Burt, 2004)
- 9.2.12 Electrical conductivity, saturated paste (Burt, 2004)
- 9.2.13 pH water (1:1, soil:water) (Burt, 2004)
- 9.2.14 pH salt (1:1, soil:0.01M CaCl₂) (Burt, 2004)
- 9.2.15 Cation Exchange Capacity, ammonium acetate, pH7
- 9.2.16 Organic matter by LOI
- 9.2.17 Bray-1 P
- 9.2.18 Mehlich-III P
- 9.2.19 Olsen P
- 9.2.20 Sodium Adsorption Ratio (SAR)

9.3 Contractor Key Personnel:

9.3.1 The Prime Contractor shall provide a Prime Contractor Representative who shall be responsible for the performance of the work and shall be the central point of contact with the Government Technical Monitor.

9.3.1.1 The name of the Prime Contractor Representative and an alternate who shall act on behalf of the Prime contractor when the Prime Contractor Representative is absent shall be designated in writing with current contact information to the Government Technical Monitor within 10 calendar days of award.

9.3.1.2 The Prime Contractor Representative or alternate shall have full authority to act for the Prime Contractor on all contract matters relating to the daily operation of this contract.

9.3.2 Only trained laboratory-trained technicians are to perform any analytical procedure.

9.4 Data Security:

9.4.1 Data subject to USDA data sharing requirements and compliance with the Geospatial Data Act.

9.4.2 Landowner PII will not be shared with contractor's laboratory.

9.4.3 The clause at FAR 52.224-3 Privacy Training will apply to this requirement.

Quality Assurance Surveillance Plan

10.0 Performance Requirement Measures/QASP:

The Government shall establish a Quality Assurance Surveillance Plan (QASP) as a means of evaluating whether the Contractor is meeting the performance standards/quality levels identified in the PWS to ensure that the government pays only for the level of services received. The requirements are stated below.

Tasks	Standards	Acceptable Quality Level (AQL)
<i>Contractor Performance Requirements and Tasks identified in Section 9</i>	<i>Standards as established in Contractor Performance Requirements and Tasks in Section 9</i>	<i>100%</i>